

Building Confidence through Progressive Learning: A Curriculum Design Strategy for New Bicycle Riders

Project Overview

Objective: Design and develop a learner-centered curriculum that enables new riders to safely build bicycle knowledge, riding skills, and confidence through a structured progression of learning activities.

Audience: Adult learners with little or no bicycle riding experience, including individuals returning to cycling after extended periods away from the activity.

Role: Lead Curriculum Designer, overseeing instructional design, assessment development, and multimedia production for the full learner journey.

Tools: Google Drive (documents, forms, drive) | Camtasia

View Live Project:

Access the [Curriculum Plan / Lesson Plan](#).

Business Challenge

Learning to ride a bicycle requires learners to simultaneously develop foundational knowledge, physical coordination, safety awareness, and confidence. Many beginner riders focus exclusively on the physical act of riding while lacking important knowledge related to bicycle components, safety equipment, and riding guidelines.

For adult learners, the challenge can be even greater. Many learners may feel embarrassed about their lack of experience and hesitate to ask questions or seek guidance. As a result, a successful learning experience needed to provide both practical skill development and foundational knowledge while creating a supportive environment that encouraged learner confidence.

This resulted in several instructional design requirements:

- Build foundational knowledge before introducing riding skills
- Reduce learner anxiety and intimidation
- Reinforce safety throughout the learning journey
- Allow learners to progress at their own pace
- Provide frequent opportunities for practice and feedback

Approach

The curriculum was designed using a progressive learning model that introduced foundational knowledge before requiring physical skill performance.

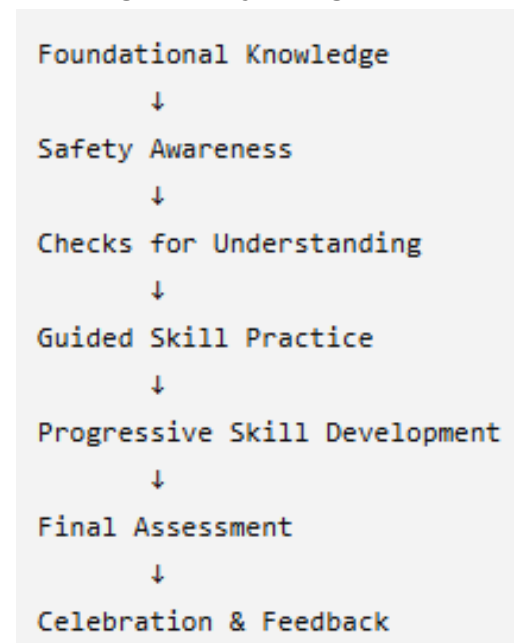
The learning journey begins with prerequisite knowledge, including bicycle anatomy, recommended riding attire, and basic riding rules. Short instructional videos and supporting resources provide learners with foundational information before they begin hands-on practice. Each knowledge section includes a check-for-understanding activity to reinforce learning and identify knowledge gaps.

After establishing foundational knowledge, learners transition into guided skill development activities. Practical exercises are organized into a sequence of increasingly complex skills, allowing learners to build confidence through incremental success. Each activity includes performance checklists and reminders to revisit prerequisite skills when additional practice is needed.

The curriculum culminates in a final assessment that reinforces both practical and conceptual knowledge. A completion celebration and learner feedback activity provide recognition for achievement while supporting continuous improvement of the curriculum.

Across the curriculum, instructional videos anchor key concepts, checks for understanding confirm retention before progression, and performance checklists guide skill practice, creating a logical, low-anxiety path from first exposure to mastery.

Learning Journey Design:



Screenshot: Bicycles - How to Ride Course Syllabus, the guide for the learning journey

- ☐ Activity 1: Watch the [Welcome Video](#).
- ☐ Activity 2: Watch the [Bicycle Anatomy Video](#) and take the [Check for Understanding](#).
- ☐ Activity 3: Watch the [Bicycle Attire Video](#) and take the [Check for Understanding](#).
- ☐ Activity 4: Study the [Rules of the Road Handout](#) and take the [Check for Understanding](#).
- ☐ Activity 5: Work through all exercises in the [Practical Exercise Handbook](#). This is the step we recommend doing with a coach or mentor for guidance and feedback.
- ☐ Activity 6: Study the [Common Challenges and Solutions Handout](#).
- ☐ Activity 7: Take the [Final Course Evaluation](#).
- ☐ Activity 8: Watch the [Celebration Video](#).
- ☐ Activity 9: Leave us [Course Feedback](#).

Key Learning & Reflection

This project sharpened my ability to sequence learning so that knowledge and skill reinforce each other rather than compete for attention. The biggest design tension was pacing: adult learners needed enough repetition to build confidence, but too much risked feeling condescending. Resolving that meant building in checkpoints learners could choose to revisit rather than forcing everyone through the same repetition. Next time, I would prototype the pacing with real learners earlier, since assumptions about "enough practice" are hard to validate on paper alone.